

Lyrics to Audio Alignment

Milestone 2 — First Version of the System

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Recap — Milestone 1

Task

Given a song's audio and its lyrics, predict a (start, end) timestamp for every lyric word.

Chosen direction (M1)

Source separation: HT Demucs (pretrained, public)

Acoustic encoder: wav2vec 2.0 (SSL pretrained on 960h speech)

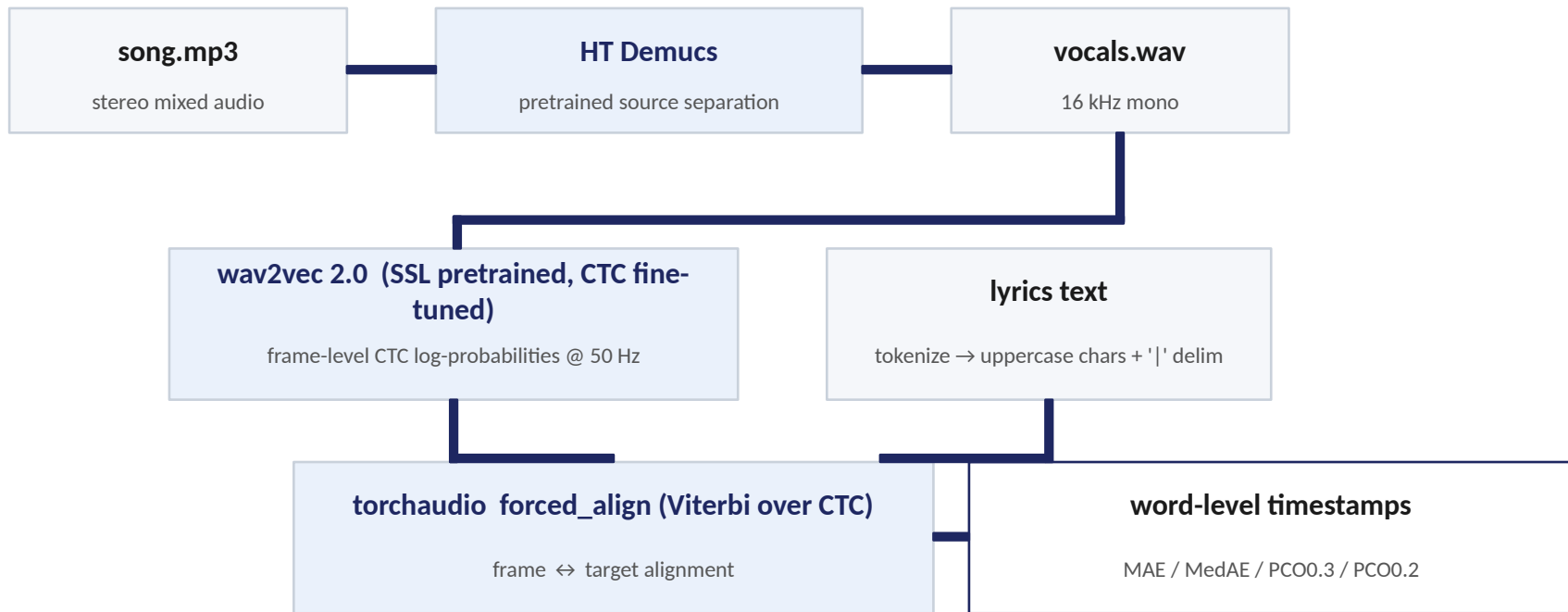
Alignment: CTC + forced alignment; masked-CE for M3

Data: evaluate on JamendoLyrics; DALI deferred to M3

This milestone (M2)

- Software architecture
- SSL component: wav2vec 2.0
- CTC + forced alignment
- Working pipeline + beta results
- Compared with literature

Software Architecture



SSL pretraining → CTC fine-tuning → forced alignment

Self-Supervised Learning — wav2vec 2.0

What is SSL here?

Pretraining objective: contrastive predictive coding on raw audio

Trained on **960 h of unlabelled LibriSpeech**

Loss: distinguish a masked-frame's quantized target from distractor frames

No transcripts needed during pretraining

Why it works for singing

- Speech and singing share phonetic structure
- Frame embeddings encode fine-grained phonetic detail
- Validated for singing by Ou et al. ISMIR 2022

facebook / wav2vec2-base-960h

Architecture: 7-layer CNN feature extractor + 12-layer Transformer

Output: frame-level log-probs over 32-token vocab

Vocabulary: uppercase A-Z, apostrophe, '|' (word boundary), <pad> (= CTC blank)

Frame rate: ~50 Hz (≈ 20 ms / frame)

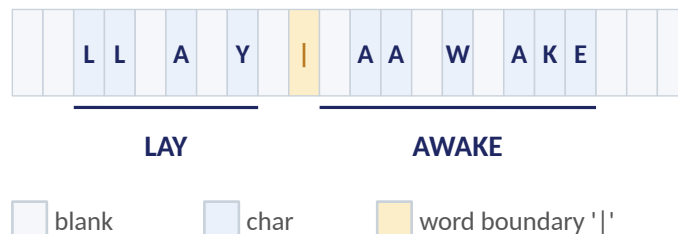
Stage we use: the released checkpoint that adds a CTC head fine-tuned on LibriSpeech

CTC Logits → Forced Alignment

How CTC turns into word timestamps

1. wav2vec 2.0 produces log-probabilities of each character per frame.
2. Tokenize the lyrics into characters with '|' between words.
3. Run torchaudio forced_align — Viterbi decoding restricted to the lyric token sequence.
4. Group consecutive frames sharing the same target token; '|' emissions become word boundaries.
5. Convert frame indices → seconds via the audio frame rate.

Frame-level CTC alignment



Milestone 3 plan: augment CTC with a masked frame-wise cross-entropy auxiliary loss (Cheng et al. 2025) using partial phoneme labels derived from word boundaries — pushes alignment precision below 0.2 s.

Tech Stack & Alternatives

Component	Chosen	Alternatives	Why we chose
Source separation	HT Demucs (Rouard 2023)	Spleeter ✓ simpler ✗ weaker quality Band-Split RNN ✓ best SDR ✗ closed-source	Open-source, transformer-based, SOTA on MUSDB
Acoustic encoder (SSL)	wav2vec 2.0 base 960h	HuBERT ✓ strong features ✗ less plug-and-play CTC MERT ✓ music-pretrained ✗ timbre, not phonetic WavLM ✓ noise-robust ✗ no singing-validated CTC	Speech CTC head ready; validated for singing (Ou 2022)
Alignment	torchaudio forced_align (Viterbi)	Greedy CTC + map ✓ simpler ✗ less precise Attention aligner ✓ flexible ✗ needs training	Optimal under CTC; deterministic; no training
Evaluation	JamendoLyrics (20 EN songs)	DALI ✓ 5k songs ✗ YouTube audio fragile In-house ✓ controlled ✗ not reproducible	Public, word-level GT, standard benchmark

Implementation Status

✓ Working in M2

- End-to-end pipeline running on M4 Max (MPS)
- HT Demucs vocal separation
- wav2vec 2.0 inference, chunked for long songs
- torchaudio.functional.forced_align
- Word-level timestamp output (JSON)
- Evaluation harness: MAE, MedAE, PCO0.3, PCO0.2
- Sample alignment visualization
- CTC greedy transcription demo

→ Planned for M3

- Fine-tune wav2vec 2.0 on singing data (DALI)
- Add masked frame-wise CE auxiliary loss (Cheng 2025)
- Replace pretrained CTC head with one trained on sung phonemes
- Multi-language evaluation on JamendoLyrics-MultiLang
- Quantitative comparison vs. published baselines on the same DALI/Jamendo split
- Robustness on songs with long instrumental intros

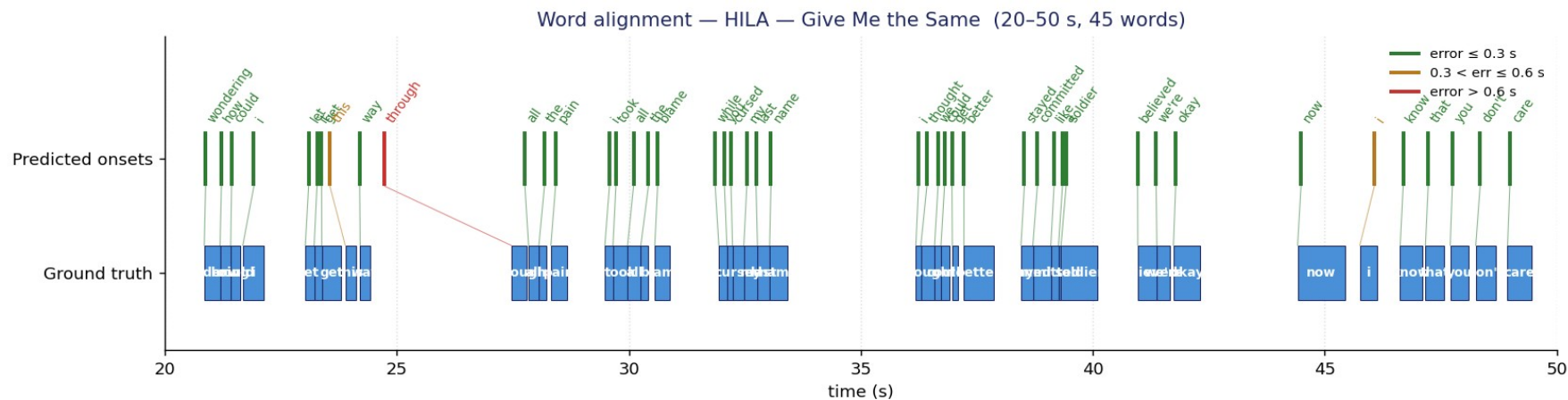
≈ 350 lines of Python — pipeline.py, run.py, visualize.py, transcribe_demo.py

Beta Results — JamendoLyrics

Configuration	Songs	MAE (s) ↓	MedAE (s) ↓	PCO 0.3 ↑	PCO 0.2 ↑
HT Demucs + wav2vec 2.0 (ours)	20	0.413	0.064	91.6%	87.0%
Raw audio + wav2vec 2.0 (no separation)	20	6.175	3.524	60.2%	56.8%
Cheng 2025 (CTC + masked CE, DALI-trained)	ref	0.216	0.041	95.2%	94.3%
Durand 2023 (contrastive, in-house data)	ref	<0.200	—	—	—

Read: Median (MedAE) is competitive even without fine-tuning. Mean (MAE) is dragged up by tail outliers — typically songs with long instrumental intros, which our M3 plan addresses.

Demo — Sample Alignment & CTC Output



First 30 seconds of HILA — Give Me the Same: predicted (top) vs. ground-truth (bottom) word boundaries

CTC greedy transcription (first 15 s of HILA vocals)

DE YOU WAKEN DI WONDERT HOW CAN NLITTLE KIS WEL RUE UP ITTLE DAY I CO

Limitations & Next Steps

Known limitations

- wav2vec 2.0 is pretrained on speech, not singing
- No fine-tuning yet — using the off-the-shelf CTC head
- Long instrumental intros mislead forced_align (e.g., Average — Embers, intro 32 s)
- English only — multilingual deferred to M3
- DALI training set not used in M2

Plan for Milestone 3

- Fine-tune wav2vec 2.0 on a DALI subset
- Add masked frame-wise CE loss (Cheng 2025)
- Voice activity detection to trim instrumental intros
- Multilingual eval on JamendoLyrics-MultiLang
- Side-by-side comparison with contrastive baseline

Thank you!

Questions?

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